# SG No. 133 $D_{4h}^{11}$ $P4_2/nbc$ [ tetragonal ]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `4a`, site symmetry: `222.`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,-2,-5,6]` |
| 2 | $[0, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,-4,7,-8]` |
| 3 | $[-X, 0, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,-10,-13,14]` |
| 4 | $[0, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,-12,15,-16]` |

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,-2,5,-6]` |
| 2 | $[-X, 0, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,-4,-7,8]` |
| 3 | $[0, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,-10,13,-14]` |
| 4 | $[X, 0, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,-12,-15,16]` |

Table 3: Wyckoff bond: `4c@4a`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,2,-5,-6]` |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,4,-7,-8]` |
| 3 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,10,-13,-14]` |
| 4 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,12,-15,-16]` |

Table 4: Wyckoff bond: `8d@4a`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,-5]` |
| 2 | $[-X, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[2,-6]` |
| 3 | $[0, X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,-8]` |
| 4 | $[0, -X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[4,-7]` |
| 5 | $[-X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,-13]` |
| 6 | $[X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[10,-14]` |
| 7 | $[0, -X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,-16]` |
| 8 | $[0, X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[12,-15]` |

Table 5: Wyckoff bond: `8e@4a`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,-6]` |
| 2 | $[0, -X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[2,-5]` |

*continued ...*

Table 5

| No. | vector | center | mapping |
|---|---|---|---|
| 3 | $[-X, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,-7]` |
| 4 | $[X, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[4,-8]` |
| 5 | $[0, -X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,-14]` |
| 6 | $[0, X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[10,-13]` |
| 7 | $[X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,-15]` |
| 8 | $[-X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[12,-16]` |

Table 6: Wyckoff bond: `8f@4a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,-4]` |
| 3 | $[-X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[5,-6]` |
| 4 | $[Y, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[7,-8]` |
| 5 | $[-X, -Y, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9,-10]` |
| 6 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11,-12]` |
| 7 | $[X, -Y, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[13,-14]` |
| 8 | $[-Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[15,-16]` |

Table 7: Wyckoff bond: `16g@4a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[16]` |

* Wyckoff site: `4b`, site symmetry: `222.`

Table 8: Wyckoff bond: `4a@4b`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,-2,-5,6]` |
| 2 | $[0, X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,-4,7,-8]` |
| 3 | $[-X, 0, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,-10,-13,14]` |
| 4 | $[0, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,-12,15,-16]` |

Table 9: Wyckoff bond: `4b@4b`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,-2,5,-6]` |
| 2 | $[-X, 0, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,-4,-7,8]` |
| 3 | $[0, -X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,-10,13,-14]` |
| 4 | $[X, 0, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,-12,-15,16]` |

Table 10: Wyckoff bond: `4c@4b`, bond symmetry: `222.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,2,-5,-6]` |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,4,-7,-8]` |
| 3 | $[0, 0, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,10,-13,-14]` |
| 4 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,12,-15,-16]` |

Table 11: Wyckoff bond: `8d@4b`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,-5]` |
| 2 | $[-X, 0, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[2,-6]` |
| 3 | $[0, X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,-8]` |
| 4 | $[0, -X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[4,-7]` |
| 5 | $[-X, 0, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,-13]` |
| 6 | $[X, 0, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[10,-14]` |
| 7 | $[0, -X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,-16]` |
| 8 | $[0, X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[12,-15]` |

Table 12: Wyckoff bond: `8e@4b`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,-6]` |
| 2 | $[0, -X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[2,-5]` |
| 3 | $[-X, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,-7]` |
| 4 | $[X, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[4,-8]` |
| 5 | $[0, -X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,-14]` |
| 6 | $[0, X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[10,-13]` |

*continued ...*

Table 12

| No. | vector | center | mapping |
|---|---|---|---|
| 7 | $[X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,-15]` |
| 8 | $[-X, 0, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[12,-16]` |

Table 13: Wyckoff bond: `8f@4b`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,-4]` |
| 3 | $[-X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[5,-6]` |
| 4 | $[Y, X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[7,-8]` |
| 5 | $[-X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9,-10]` |
| 6 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11,-12]` |
| 7 | $[X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[13,-14]` |
| 8 | $[-Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[15,-16]` |

Table 14: Wyckoff bond: `16g@4b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[16]` |

* Wyckoff site: `4c`, site symmetry: `2.22`

Table 15: Wyckoff bond: `4a@4c`, bond symmetry: `2.22`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,-2,7,-8]` |
| 2 | $[-X, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[3,-4,5,-6]` |
| 3 | $[-X, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[9,-10,15,-16]` |
| 4 | $[X, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[11,-12,13,-14]` |

Table 16: Wyckoff bond: `4b@4c`, bond symmetry: `2.22`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, -X, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,-2,-7,8]` |
| 2 | $[X, X, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3,-4,-5,6]` |
| 3 | $[-X, X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9,-10,-15,16]` |
| 4 | $[-X, -X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11,-12,-13,14]` |

Table 17: Wyckoff bond: `4c@4c`, bond symmetry: `2.22`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,2,-7,-8]` |
| 2 | $[0, 0, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3,4,-5,-6]` |
| 3 | $[0, 0, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9,10,-15,-16]` |
| 4 | $[0, 0, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11,12,-13,-14]` |

Table 18: Wyckoff bond: `8d@4c`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,-8]` |
| 2 | $[-X, -X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[2,-7]` |
| 3 | $[-X, X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3,-6]` |
| 4 | $[X, -X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[4,-5]` |
| 5 | $[-X, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9,-16]` |
| 6 | $[X, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[10,-15]` |
| 7 | $[X, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11,-14]` |
| 8 | $[-X, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[12,-13]` |

Table 19: Wyckoff bond: `8e@4c`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, -X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,-7]` |
| 2 | $[-X, X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[2,-8]` |
| 3 | $[X, X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3,-5]` |
| 4 | $[-X, -X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[4,-6]` |
| 5 | $[-X, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9,-15]` |
| 6 | $[X, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[10,-16]` |
| 7 | $[-X, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11,-13]` |
| 8 | $[X, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[12,-14]` |

Table 20: Wyckoff bond: `8f@4c`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3,-4]` |

*continued ...*

Table 20

| No. | vector | center | mapping |
|---|---|---|---|
| 3 | $[-X, Y, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[5,-6]` |
| 4 | $[Y, X, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[7,-8]` |
| 5 | $[-X, -Y, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9,-10]` |
| 6 | $[Y, -X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11,-12]` |
| 7 | $[X, -Y, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[13,-14]` |
| 8 | $[-Y, -X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[15,-16]` |

Table 21: Wyckoff bond: `16g@4c`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[16]` |

* Wyckoff site: `4d`, site symmetry: `-4..`

Table 22: Wyckoff bond: `4a@4d`, bond symmetry: `-4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[1,2,-11,-12]` |
| 2 | $[0, 0, Z]$ | $[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[3,4,-9,-10]` |
| 3 | $[0, 0, -Z]$ | $[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[5,6,-15,-16]` |
| 4 | $[0, 0, -Z]$ | $[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[7,8,-13,-14]` |

Table 23: Wyckoff bond: `8b@4d`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$ | `[3,-4]` |
| 3 | $[-X, Y, 0]$ | $[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[5,-6]` |
| 4 | $[Y, X, 0]$ | $[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[7,-8]` |

*continued ...*

Table 23

| No. | vector | center | mapping |
|---|---|---|---|
| 5 | $[-X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[9,-10]` |
| 6 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[11,-12]` |
| 7 | $[X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[13,-14]` |
| 8 | $[-Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[15,-16]` |

Table 24: Wyckoff bond: `16c@4d`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[16]` |

* Wyckoff site: `8e`, site symmetry: `-1`

Table 25: Wyckoff bond: `8a@8e`, bond symmetry: `-1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[0, 0, 0]$ | `[1,-9]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[2,-10]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[3,-11]` |
| 4 | $[Y, -X, Z]$ | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[4,-12]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{1}{2}, 0, 0\right]$ | `[5,-13]` |
| 6 | $[X, -Y, -Z]$ | $\left[0, \frac{1}{2}, 0\right]$ | `[6,-14]` |
| 7 | $[Y, X, -Z]$ | $\left[0, 0, \frac{1}{2}\right]$ | `[7,-15]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[8,-16]` |

* Wyckoff site: `8f`, site symmetry: `2..`

Table 26: Wyckoff bond: `8a@8f`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | `[3,-4]` |

*continued ...*

Table 26

| No. | vector | center | mapping |
|---|---|---|---|
| 3 | $[-X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, -z\right]$ | [5,-6] |
| 4 | $[Y, X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [7,-8] |
| 5 | $[-X, -Y, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | [9,-10] |
| 6 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [11,-12] |
| 7 | $[X, -Y, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, z\right]$ | [13,-14] |
| 8 | $[-Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [15,-16] |

Table 27: Wyckoff bond: 8b@8f, bond symmetry: 2..

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | [1,2] |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | [3,4] |
| 3 | $[0, 0, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, -z\right]$ | [5,6] |
| 4 | $[0, 0, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [7,8] |
| 5 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | [9,10] |
| 6 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [11,12] |
| 7 | $[0, 0, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z\right]$ | [13,14] |
| 8 | $[0, 0, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [15,16] |

Table 28: Wyckoff bond: 16c@8f, bond symmetry: 1

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | [1] |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | [2] |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | [3] |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | [4] |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, -z\right]$ | [5] |
| 6 | $[X, -Y, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, -z\right]$ | [6] |
| 7 | $[Y, X, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [7] |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [8] |
| 9 | $[-X, -Y, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | [9] |
| 10 | $[X, Y, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | [10] |
| 11 | $[Y, -X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [11] |
| 12 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [12] |
| 13 | $[X, -Y, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z\right]$ | [13] |
| 14 | $[-X, Y, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z\right]$ | [14] |
| 15 | $[-Y, -X, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [15] |
| 16 | $[Y, X, Z]$ | $\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [16] |

* Wyckoff site: 8g, site symmetry: 2..

Table 29: Wyckoff bond: `8a@8g`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[3,-4]` |
| 3 | $[-X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | `[5,-6]` |
| 4 | $[Y, X, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[7,-8]` |
| 5 | $[-X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[9,-10]` |
| 6 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[11,-12]` |
| 7 | $[X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[13,-14]` |
| 8 | $[-Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[15,-16]` |

Table 30: Wyckoff bond: `8b@8g`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1,2]` |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[3,4]` |
| 3 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | `[5,6]` |
| 4 | $[0, 0, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[7,8]` |
| 5 | $[0, 0, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[9,10]` |
| 6 | $[0, 0, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[11,12]` |
| 7 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[13,14]` |
| 8 | $[0, 0, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[15,16]` |

Table 31: Wyckoff bond: `16c@8g`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[16]` |

* Wyckoff site: `8h`, site symmetry: `.2.`

Table 32: Wyckoff bond: `8a@8h`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0,\ X,\ Z]$ | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[1,-6]` |
| 2 | $[0,\ -X,\ Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$ | `[2,-5]` |
| 3 | $[-X,\ 0,\ Z]$ | $\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$ | `[3,-7]` |
| 4 | $[X,\ 0,\ Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[4,-8]` |
| 5 | $[0,\ -X,\ -Z]$ | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[9,-14]` |
| 6 | $[0,\ X,\ -Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[10,-13]` |
| 7 | $[X,\ 0,\ -Z]$ | $\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$ | `[11,-15]` |
| 8 | $[-X,\ 0,\ -Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[12,-16]` |

Table 33: Wyckoff bond: `8b@8h`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\ 0,\ 0]$ | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[1,6]` |
| 2 | $[-X,\ 0,\ 0]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$ | `[2,5]` |
| 3 | $[0,\ X,\ 0]$ | $\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$ | `[3,7]` |
| 4 | $[0,\ -X,\ 0]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[4,8]` |
| 5 | $[-X,\ 0,\ 0]$ | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[9,14]` |
| 6 | $[X,\ 0,\ 0]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[10,13]` |
| 7 | $[0,\ -X,\ 0]$ | $\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$ | `[11,15]` |
| 8 | $[0,\ X,\ 0]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[12,16]` |

Table 34: Wyckoff bond: `16c@8h`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\ Y,\ Z]$ | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[1]` |
| 2 | $[-X,\ -Y,\ Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$ | `[2]` |
| 3 | $[-Y,\ X,\ Z]$ | $\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$ | `[3]` |
| 4 | $[Y,\ -X,\ Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[4]` |
| 5 | $[-X,\ Y,\ -Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$ | `[5]` |
| 6 | $[X,\ -Y,\ -Z]$ | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[6]` |
| 7 | $[Y,\ X,\ -Z]$ | $\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$ | `[7]` |
| 8 | $[-Y,\ -X,\ -Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[8]` |
| 9 | $[-X,\ -Y,\ -Z]$ | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[9]` |
| 10 | $[X,\ Y,\ -Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[10]` |
| 11 | $[Y,\ -X,\ -Z]$ | $\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$ | `[11]` |
| 12 | $[-Y,\ X,\ -Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[12]` |
| 13 | $[X,\ -Y,\ Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[13]` |
| 14 | $[-X,\ Y,\ Z]$ | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[14]` |
| 15 | $[-Y,\ -X,\ Z]$ | $\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$ | `[15]` |
| 16 | $[Y,\ X,\ Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[16]` |

* Wyckoff site: `8i`, site symmetry: `.2.`

Table 35: Wyckoff bond: `8a@8i`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0,\ X,\ Z]$ | $\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[1,-6]` |
| 2 | $[0,\ -X,\ Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[2,-5]` |
| 3 | $[-X,\ 0,\ Z]$ | $\left[\frac{1}{4},\ x,\ 0\right]$ | `[3,-7]` |
| 4 | $[X,\ 0,\ Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$ | `[4,-8]` |
| 5 | $[0,\ -X,\ -Z]$ | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[9,-14]` |
| 6 | $[0,\ X,\ -Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[10,-13]` |
| 7 | $[X,\ 0,\ -Z]$ | $\left[\frac{3}{4},\ -x,\ 0\right]$ | `[11,-15]` |
| 8 | $[-X,\ 0,\ -Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$ | `[12,-16]` |

Table 36: Wyckoff bond: `8b@8i`, bond symmetry: `.2.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\ 0,\ 0]$ | $\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[1,6]` |
| 2 | $[-X,\ 0,\ 0]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[2,5]` |
| 3 | $[0,\ X,\ 0]$ | $\left[\frac{1}{4},\ x,\ 0\right]$ | `[3,7]` |
| 4 | $[0,\ -X,\ 0]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$ | `[4,8]` |
| 5 | $[-X,\ 0,\ 0]$ | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[9,14]` |
| 6 | $[X,\ 0,\ 0]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[10,13]` |
| 7 | $[0,\ -X,\ 0]$ | $\left[\frac{3}{4},\ -x,\ 0\right]$ | `[11,15]` |
| 8 | $[0,\ X,\ 0]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$ | `[12,16]` |

Table 37: Wyckoff bond: `16c@8i`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\ Y,\ Z]$ | $\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[1]` |
| 2 | $[-X,\ -Y,\ Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[2]` |
| 3 | $[-Y,\ X,\ Z]$ | $\left[\frac{1}{4},\ x,\ 0\right]$ | `[3]` |
| 4 | $[Y,\ -X,\ Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$ | `[4]` |
| 5 | $[-X,\ Y,\ -Z]$ | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[5]` |
| 6 | $[X,\ -Y,\ -Z]$ | $\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[6]` |
| 7 | $[Y,\ X,\ -Z]$ | $\left[\frac{1}{4},\ x,\ 0\right]$ | `[7]` |
| 8 | $[-Y,\ -X,\ -Z]$ | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$ | `[8]` |
| 9 | $[-X,\ -Y,\ -Z]$ | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[9]` |
| 10 | $[X,\ Y,\ -Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[10]` |
| 11 | $[Y,\ -X,\ -Z]$ | $\left[\frac{3}{4},\ -x,\ 0\right]$ | `[11]` |
| 12 | $[-Y,\ X,\ -Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$ | `[12]` |
| 13 | $[X,\ -Y,\ Z]$ | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[13]` |
| 14 | $[-X,\ Y,\ Z]$ | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[14]` |
| 15 | $[-Y,\ -X,\ Z]$ | $\left[\frac{3}{4},\ -x,\ 0\right]$ | `[15]` |
| 16 | $[Y,\ X,\ Z]$ | $\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$ | `[16]` |

* Wyckoff site: `8j`, site symmetry: `..2`

Table 38: Wyckoff bond: `8a@8j`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, -X, Z]$ | $\left[x, x, \frac{1}{4}\right]$ | `[1,-7]` |
| 2 | $[-X, X, Z]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{4}\right]$ | `[2,-8]` |
| 3 | $[X, X, Z]$ | $\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$ | `[3,-5]` |
| 4 | $[-X, -X, Z]$ | $\left[x, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[4,-6]` |
| 5 | $[-X, X, -Z]$ | $\left[-x, -x, \frac{3}{4}\right]$ | `[9,-15]` |
| 6 | $[X, -X, -Z]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[10,-16]` |
| 7 | $[-X, -X, -Z]$ | $\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$ | `[11,-13]` |
| 8 | $[X, X, -Z]$ | $\left[-x, x+\frac{1}{2}, \frac{1}{4}\right]$ | `[12,-14]` |

Table 39: Wyckoff bond: `8b@8j`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, X, 0]$ | $\left[x, x, \frac{1}{4}\right]$ | `[1,7]` |
| 2 | $[-X, -X, 0]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{4}\right]$ | `[2,8]` |
| 3 | $[-X, X, 0]$ | $\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$ | `[3,5]` |
| 4 | $[X, -X, 0]$ | $\left[x, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[4,6]` |
| 5 | $[-X, -X, 0]$ | $\left[-x, -x, \frac{3}{4}\right]$ | `[9,15]` |
| 6 | $[X, X, 0]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[10,16]` |
| 7 | $[X, -X, 0]$ | $\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$ | `[11,13]` |
| 8 | $[-X, X, 0]$ | $\left[-x, x+\frac{1}{2}, \frac{1}{4}\right]$ | `[12,14]` |

Table 40: Wyckoff bond: `16c@8j`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[x, x, \frac{1}{4}\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{4}\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[x, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[x, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[x, x, \frac{1}{4}\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{4}\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $\left[-x, -x, \frac{3}{4}\right]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[-x, x+\frac{1}{2}, \frac{1}{4}\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[-x, x+\frac{1}{2}, \frac{1}{4}\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[-x, -x, \frac{3}{4}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[16]` |

* Wyckoff site: `16k`, site symmetry: `1`

Table 41: Wyckoff bond: `16a@16k`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{2} - x, \frac{1}{2} - y, z\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{2} - y, x, z + \frac{1}{2}\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[y, \frac{1}{2} - x, z + \frac{1}{2}\right]$ | `[4]` |
| 5 | $[-X, Y, -Z]$ | $\left[\frac{1}{2} - x, y, -z\right]$ | `[5]` |
| 6 | $[X, -Y, -Z]$ | $\left[x, \frac{1}{2} - y, -z\right]$ | `[6]` |
| 7 | $[Y, X, -Z]$ | $\left[y, x, \frac{1}{2} - z\right]$ | `[7]` |
| 8 | $[-Y, -X, -Z]$ | $\left[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2} - z\right]$ | `[8]` |
| 9 | $[-X, -Y, -Z]$ | $[-x, -y, -z]$ | `[9]` |
| 10 | $[X, Y, -Z]$ | $\left[x + \frac{1}{2}, y + \frac{1}{2}, -z\right]$ | `[10]` |
| 11 | $[Y, -X, -Z]$ | $\left[y + \frac{1}{2}, -x, \frac{1}{2} - z\right]$ | `[11]` |
| 12 | $[-Y, X, -Z]$ | $\left[-y, x + \frac{1}{2}, \frac{1}{2} - z\right]$ | `[12]` |
| 13 | $[X, -Y, Z]$ | $\left[x + \frac{1}{2}, -y, z\right]$ | `[13]` |
| 14 | $[-X, Y, Z]$ | $\left[-x, y + \frac{1}{2}, z\right]$ | `[14]` |
| 15 | $[-Y, -X, Z]$ | $\left[-y, -x, z + \frac{1}{2}\right]$ | `[15]` |
| 16 | $[Y, X, Z]$ | $\left[y + \frac{1}{2}, x + \frac{1}{2}, z + \frac{1}{2}\right]$ | `[16]` |